

# Regime Structure in Adaptive Tariff Conflicts: A Multi-Agent Reinforcement Learning Analysis

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**Abstract**—Tariff conflicts unfold over time as adaptive, history-dependent policy exchanges, a structure that closed-form Stackelberg and one-shot optimal-tariff models capture only imperfectly. We study a two-country trade conflict in which a structurally privileged country deploys tariffs and currency depreciation and a responding country sets counter-tariffs and competitive depreciation, with both agents updating policies simultaneously rather than under explicit Stackelberg commitment. Agents learn via tabular Q-learning, with the follower using double-Q updates to reduce overestimation bias on its larger joint action space; this yields co-adaptive policies without imposing equilibrium assumptions and keeps learned value functions directly inspectable. Elasticities and policy ranges are calibrated to the 2018–2019 US–China episode (import demand elasticities of 1.3–1.55, tariff ceilings of 25%–35%), with results reported across an  $8 \times 8$  structural parameter grid and five seeds per cell. We identify three stable outcome regimes (Deterrence, Transition, Escalation) and three findings. First, the Deterrence–Escalation boundary tracks a compact structural predictor: the ratio of the follower’s retaliation ceiling to the leader’s tariff ceiling. Second, leader currency depreciation extends Deterrence only at moderate intensity, with an interior optimum beyond which inflation costs dominate competitiveness gains. Third, diplomatic cost and export sensitivity jointly compress the Deterrence region. Near regime boundaries, identical parameters produce divergent stable outcomes across seeds, indicating coordination-sensitive regions with multiple attractors rather than learning noise; sensitivity to Q-initialization confirms this. The phase-diagram methodology generalizes to other repeated games whose outcome type depends sensitively on structural parameters.

**Index Terms**—Multi-agent reinforcement learning; trade conflict; regime structure; phase-diagram analysis; Q-learning

## I. INTRODUCTION

Trade conflicts between major economies involve sequential, adaptive decision-making across multiple policy instruments, including tariffs, retaliation, and currency adjustments, that resist simple characterisation. The 2018–2019 US–China dispute illustrates this clearly: tariff rounds were announced, countered, and escalated over months, with each side conditioning its actions on the other’s prior moves. Empirical studies of that episode suggest import demand elasticities broadly in the range of 1–2.5 and document average tariff increases on the order of 16–25%, with higher rates in targeted sectors [1],

[7]. These estimates establish the empirical envelope within which trade conflict models should operate, but they do not themselves explain when escalation occurs, when deterrence holds, or what structural conditions determine the outcome.

Standard approaches fall into two broad categories. Optimal tariff theory characterises best-response tariff levels under various market structures but typically assumes static or simultaneous play. Game-theoretic extensions, including Stackelberg leadership models [9] in which one country moves first and the other responds, solve for closed-form equilibria under full rationality and common knowledge assumptions. These frameworks yield important insights into long-run trade policy incentives [2], [3] but are less suited to situations where strategies co-evolve over time, retaliation thresholds are uncertain, and the conflict may settle into one of several possible outcomes depending on the path taken.

Multi-agent reinforcement learning (MARL) offers a complementary approach. Rather than imposing equilibrium assumptions, MARL allows agents to discover strategies through repeated interaction, making it well suited to settings where policy responses are adaptive and history-dependent. Tabular Q-learning in particular offers interpretability advantages over deep RL alternatives: learned value functions are directly inspectable and convergence properties are well-characterised [13]. Recent work has applied Q-learning to oligopoly pricing [4] and Cournot competition [10], but its application to trade conflict with asymmetric strategic roles and currency instruments remains limited.

In this paper we model a two-country tariff conflict as a repeated game with asymmetric strategic roles and simulate it using tabular Q-learning across a structured parameter grid. Parameter values are selected as representative points within empirically observed ranges to reflect contrasting structural regimes, rather than direct point estimates of any specific episode. The central output is a set of phase diagrams mapping structural parameters, including retaliation capacity, tariff space, and depreciation range, to three stable outcome regimes: Deterrence, Transition, and Escalation. This phase-diagram methodology reveals not just realized outcomes, but

the structural conditions that determine which regime a conflict enters.

Our main contributions are:

- **Regime boundary identification:** The Deterrence–Escalation boundary is primarily determined by the ratio of the follower’s retaliation capacity to the leader’s tariff space, providing a compact, interpretable predictor of conflict outcome across the parameter grid.
- **Depreciation effects on the regime boundary:** Moderate leader currency depreciation shifts the Deterrence boundary, though the effect is non-monotonic. At higher levels, depreciation introduces coordination-sensitive regions rather than extending deterrence.
- **Coordination sensitivity:** Near structural boundaries, identical parameters produce divergent outcomes across learning runs, revealing regions with multiple stable policy regimes. This reflects underlying coordination sensitivity rather than noise in the learning process.

The remainder of the paper is organised as follows. Section II presents the economic model and calibration. Section III describes the learning architecture and experimental setup. Section IV reports results across the phase diagram suite. Section V discusses economic and methodological implications. Section VI addresses limitations. Section VII concludes.

## II. MODEL

We model a two-country trade conflict as a discrete-time game with asymmetric strategic roles. The leader represents a large economy with the capacity to set tariffs and deploy currency depreciation as a strategic instrument. The follower responds each period with its own depreciation rate and a retaliatory tariff. Both countries trade a single composite good, and trade flows adjust gradually to policy changes through partial adjustment dynamics. Behavioural parameters for inequity aversion, spite, and diplomatic costs are set to zero throughout, so equilibrium outcomes are driven entirely by structural parameters.

### A. Trade Flows

Three trade flow variables are central to the model: imports demanded by the leader ( $M$ ), exports supplied by the follower ( $X$ ), and exports from the leader ( $E$ ). Each responds to prevailing policy instruments through standard elasticity relationships.

Import demand falls with the leader’s tariff rate  $\tau$ :

$$M^* = M_0 \cdot (1 + \tau)^{-\varepsilon_d} \quad (1)$$

where  $\varepsilon_d$  is the import demand elasticity and  $M_0$  is the baseline import level. A higher tariff suppresses import volume, generating tariff revenue for the leader while reducing consumer welfare.

Follower export supply rises with the follower’s depreciation rate  $d$ :

$$X^* = X_0 \cdot (1 + d)^{\varepsilon_s} \quad (2)$$

where  $\varepsilon_s$  is the export supply elasticity. Depreciation makes the follower’s goods cheaper relative to the leader’s market, boosting export volume at the cost of domestic inflation.

Leader export demand depends on both the follower’s retaliatory tariff  $\rho$  and the leader’s own depreciation  $d_L$ :

$$E^* = E_0 \cdot (1 + \rho)^{-\varepsilon_E} \cdot (1 + d_L)^{\varepsilon_{dL}} \quad (3)$$

Retaliation suppresses demand for leader exports, while leader depreciation partially offsets this by improving price competitiveness. The depreciation elasticity  $\varepsilon_{dL}$  is set to 0.60, consistent with empirical estimates showing that exchange-rate effects on export volumes are typically below unity and vary systematically across products, with larger responses for less sophisticated goods [5]. This parameter is held fixed across all calibrations.

Cross-channel elasticities linking tariffs to follower exports and depreciation to leader exports are set to zero in the baseline specification to isolate the primary strategic channels.

Trade flows do not adjust instantaneously. Each variable follows a partial adjustment process toward its target level:

$$M_t = (1 - \kappa) M_{t-1} + \kappa M^* \quad (4)$$

$$X_t = (1 - \lambda) X_{t-1} + \lambda X^* \quad (5)$$

$$E_t = (1 - \eta) E_{t-1} + \eta E^* \quad (6)$$

where  $\kappa$ ,  $\lambda$ , and  $\eta$  are the adjustment speeds for imports, follower exports, and leader exports respectively. Lower values imply greater inertia, so trade flows respond gradually rather than instantaneously to policy changes.

### B. Payoff Functions

Each agent receives a per-period payoff that reflects the economic costs and benefits of its policy choices.

The leader’s payoff is:

$$\begin{aligned} L = & \tau M - 0.5 \tau (M_0 - M) - c_L \tau^2 \\ & - \psi_E \max(0, E_0 - E) - \phi_L d_L I_{L0} - c_{dL} d_L^2 \\ & - \xi \left( \frac{\rho}{\rho_{\max}} \right) (\tau M) \end{aligned} \quad (7)$$

The components are as follows. Tariff revenue increases with  $\tau$  and import volume. The consumer surplus loss term scales with both the tariff rate and the reduction in imports. The quadratic term  $c_L \tau^2$  captures increasing political and economic costs of high tariffs. The term  $\psi_E$  penalises export losses below baseline, reflecting the leader’s sensitivity to retaliation.

The next two terms capture the cost of depreciation. The linear term  $\phi_L d_L I_{L0}$  represents inflationary pressure arising from higher import prices following depreciation, scaled by the leader’s baseline import exposure  $I_{L0}$ . The quadratic term  $c_{dL} d_L^2$  captures increasing implementation and macroeconomic costs of aggressive currency intervention.

In addition to these economic components, we incorporate an external diplomatic cost arising from retaliation. We model this as  $\xi (\rho / \rho_{\max}) (\tau M)$ , motivated by the terms-of-trade theory of trade agreements in [2], where retaliation

serves as an enforcement mechanism that disciplines unilateral tariff distortions. In this framework, governments have an incentive to raise tariffs to shift costs onto trading partners, and trade agreements rely on the threat of retaliation to sustain cooperative outcomes. The term  $\tau M$  captures the scale of the tariff distortion, while  $\rho/\rho_{\max}$  reflects the intensity of the follower's response relative to its feasible retaliation capacity. The parameter  $\xi$  summarizes the resulting political and diplomatic costs to the leader. The multiplicative form ensures these costs increase jointly with the magnitude of the distortion and the strength of retaliation.

This term is conceptually distinct from  $c_L \tau^2$ , which captures domestic political and implementation costs of tariff policy. In contrast,  $\xi$  reflects external costs arising from retaliation and broader diplomatic consequences.

In the baseline experiments, we set  $\xi = 0$ , so equilibrium outcomes are driven entirely by structural parameters. The diplomatic cost term is introduced only in the perturbation analysis of Section IV-E.

The follower's payoff is:

$$F = (X - X_0) - \phi_F d I_{F0} - c_F d^2 - c_\rho \rho^2 \quad (8)$$

The first term represents export gains relative to baseline. The second and third terms capture the inflationary and quadratic costs of depreciation. The linear term  $\phi_F d I_{F0}$  represents inflationary pressure scaled by the follower's baseline import exposure  $I_{F0}$ , reflecting domestic price pressures arising from currency depreciation. The final term  $c_\rho \rho^2$  reflects the cost of retaliation, capturing political friction, supply chain disruption, and welfare losses from import suppression.

Crucially,  $\rho$  does not directly increase follower export revenue; its role is purely strategic, operating through its effect on leader export losses and the subsequent leader response. This asymmetry has important implications for follower learning dynamics, discussed in Section IV-G.

For comparability across agents, we set  $I_{L0} = I_{F0} = M_0$  in the baseline calibration, so that differences in outcomes are driven by strategic interactions rather than differences in import exposure. Allowing for asymmetric import exposure is a natural extension and would alter the relative strength of inflationary channels without changing the qualitative structure of the game.

### C. Calibration

The model is simulated under three parameter configurations, summarised in Table I. The Leader-Favoring (LF) calibration reflects a structural environment in which the leader holds relatively greater policy capacity, with a higher tariff ceiling, lower retaliation range, slower import adjustment, and elasticities near the lower empirical bound. The Follower-Favoring (FF) calibration reflects the opposite case, with a higher retaliation ceiling, faster adjustment speeds, stronger export supply response, and elasticities near the upper empirical range. The CENTER calibration lies near the structural Deterrence–Escalation boundary by design and serves as a sensitivity reference point.

TABLE I  
MODEL CALIBRATIONS: LF, FF, AND CENTER

| Parameter (Symbol)                       | LF   | FF   | CENTER |
|------------------------------------------|------|------|--------|
| Import demand elast. ( $\varepsilon_d$ ) | 1.30 | 1.55 | 1.42   |
| Export supply elast. ( $\varepsilon_s$ ) | 0.85 | 1.20 | 1.00   |
| Leader export elast. ( $\varepsilon_E$ ) | 1.00 | 1.50 | 1.25   |
| Import adj. speed ( $\kappa$ )           | 0.20 | 0.60 | 0.60   |
| Export adj. speed ( $\lambda$ )          | 0.40 | 0.55 | 0.55   |
| Leader export adj. speed ( $\eta$ )      | 0.30 | 0.55 | 0.55   |
| Tariff ceiling ( $\tau_{\max}$ )         | 0.35 | 0.25 | 0.28   |
| Follower depr. ceiling ( $d_{\max}$ )    | 0.18 | 0.25 | 0.22   |
| Retaliation ceiling ( $\rho_{\max}$ )    | 0.20 | 0.35 | 0.27   |
| Export loss sensitivity ( $\psi_E$ )     | 0.90 | 1.40 | 1.15   |
| Leader depr. ceiling ( $d_{L,\max}$ )    | 0.15 | 0.10 | 0.12   |
| Leader inflation cost ( $\phi_L$ )       | 0.30 | 0.30 | 0.30   |

Parameter values are selected as representative calibration points within empirically observed ranges to reflect contrasting structural regimes (LF: leader-favoring, FF: follower-favoring), rather than direct point estimates. Import demand elasticities of 1.30–1.55 and tariff ceilings of 25%–35% are consistent with empirical estimates from the 2018–2019 US–China trade conflict [1], [7].

All three calibration points appear as reference markers on every phase diagram, allowing results to be interpreted relative to the structural position of each regime.

## III. LEARNING AND EXPERIMENTAL SETUP

### A. Tabular Q-Learning

Both agents learn via tabular Q-learning with epsilon-greedy exploration. We adopt tabular rather than deep reinforcement learning methods for two reasons. First, the discrete action and state spaces are sufficiently small that tabular methods exhibit stable convergence in this setting without function approximation. Second, tabular Q-functions are directly inspectable, making it possible to verify convergence and diagnose learning pathologies, which is important when the goal is to characterise equilibrium structure rather than maximise performance.

Each agent maintains a Q-table mapping state–action pairs to estimated values. At each step, the agent selects the greedy action with probability  $1 - \varepsilon$  and a random action with probability  $\varepsilon$ , where  $\varepsilon$  here denotes the exploration rate (distinct from the elasticity parameters of Section II). Q-values are updated according to the standard rule [8], [13]:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (9)$$

where  $r$  is the per-period payoff,  $\gamma = 0.98$  corresponds to the model discount factor  $\delta$ , and  $\alpha$  is the learning rate. Learning rates are adaptive, decaying as  $\max(\alpha_{\min}, 1/\sqrt{n})$ , where  $n$  is the visit count for state–action pair  $(s, a)$ . This ensures early updates are responsive while later updates stabilise.

The leader uses single-Q updates with  $\alpha_{\min} = 0.02$ . The follower uses double-Q updates with  $\alpha_{\min} = 0.05$ , as described in Section III-B. Q-values are initialised to 3000 throughout, implementing optimistic initialisation that encourages broad early exploration before agents begin to exploit learned policies.

### B. Follower Learning: Double Q-Learning

The follower learns over a joint action space  $(d, \rho)$  using tabular Q-learning. To reduce overestimation bias in action-value estimates, the follower employs Double Q-learning [11], [12]. Two value functions,  $Q_1$  and  $Q_2$ , are maintained and updated symmetrically. At each update step, one of the tables is selected at random for update, while the other is used for evaluation.

When updating  $Q_1$ , the target is:

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha[r + \gamma Q_2(s', \arg \max_{a'} Q_1(s', a')) - Q_1(s, a)] \quad (10)$$

Similarly, when updating  $Q_2$ , the target is:

$$Q_2(s, a) \leftarrow Q_2(s, a) + \alpha[r + \gamma Q_1(s', \arg \max_{a'} Q_2(s', a')) - Q_2(s, a)] \quad (11)$$

This decouples action selection from evaluation, mitigating the positive bias that arises when both are performed using the same value function.

The leader uses standard single Q-learning. Overestimation bias in the leader's value function is less critical, as the leader's optimal policy is generally monotonic in the tariff dimension within a given regime. In contrast, the follower faces a larger joint action space over  $(d, \rho)$ , where overestimation bias can distort relative valuations across instruments. Double Q-learning stabilises these comparisons by reducing bias in action evaluation, leading to more balanced responses across depreciation and retaliation.

Both agents observe a discretised representation of recent policy history and current trade flow conditions. State variables are binned into a small number of discrete levels, ensuring that the overall state space remains tractable for tabular learning.

The leader's state  $(d_1, d_2, d_3, \tau_{\text{last}}, M_{\text{bin}}, X_{\text{bin}}, \rho_{\text{last}})$  is a 7-tuple which encodes the follower's three most recent depreciation actions, the leader's most recent tariff, binned import and export levels, and the follower's last retaliation level. When leader depreciation is active ( $d_{L, \text{max}} > 0$ ), an additional element  $d_{L, \text{bin}}$  is appended. The leader's action space consists of uniformly spaced tariff bins over  $[0, \tau_{\text{max}}]$  in the baseline specification; when depreciation is active, the action space expands to joint  $(\tau, d_L)$  choices.

The follower's state  $(\tau_3, \tau_2, \tau_1, X_{\text{bin}}, d_{\text{last}}, \rho_{\text{last}}, X_{\text{momentum}})$  is a 7-tuple which encodes the leader's recent tariff history, the current export level, the follower's own last actions, and a simple export momentum signal. When leader depreciation is active,  $d_{L, \text{bin}}$  is included as an additional state variable. The follower's joint action space contains at most 30  $(d, \rho)$  combinations in the baseline configuration, ensuring tractable tabular learning.

### C. Experimental Setup

Simulations are conducted on an  $8 \times 8$  parameter grid for each phase diagram. Each grid point corresponds to a fixed pair of structural parameters and is evaluated across 5 independent

seeds with different random initialisations. Training runs for 600,000 steps per seed in the baseline configuration and 1,000,000 steps in the depreciation extension, reflecting the larger joint action space when leader depreciation is included. Exploration decays from  $\varepsilon = 0.15$  to  $\varepsilon = 0.02$  over the training horizon. All reported results are based on behaviour in the final evaluation window, defined as the last 20% of training steps.

### D. Regime Classification

At the end of each run, the mean leader payoff over the final evaluation window is used to classify the outcome into one of three regimes: *Deterrence* (mean payoff  $> 15$ ), *Transition* (mean payoff in  $[-15, 15]$ ), or *Escalation* (mean payoff  $< -15$ ). The final regime assigned to each grid point is determined by strict majority vote across the five seeds. If no regime achieves a majority, the grid point is classified as *Multiple*.

The Multiple classification does not reflect a failure of convergence. Within each seed, agents converge to stable policies. Disagreement across seeds arises because, near structural boundaries, the system admits multiple stable policy regimes. Which regime is realised depends on the path taken during early exploration.

### E. Initialization Sensitivity

The primary phase diagrams are generated using Q-initialisation of 3000 for both agents. As a robustness check, selected grid points are re-evaluated with lower initialisation for the follower. Lower initialisation reduces the incentive for optimistic exploration, and grid points that appear as clean Deterrence or Escalation under high initialisation often resolve into Multiple classifications. This result strengthens the interpretation of the Multiple regime: the overall phase boundary remains stable across initialisations, confirming that regime separation is structurally determined, while the sensitivity to initial conditions reflects genuine coordination dynamics.

## IV. RESULTS

### A. Overview

Across all experiments, the system converges to one of three stable regimes: Deterrence, Transition, or Escalation, organised into distinct phase regions separated by relatively sharp boundaries. A fourth outcome, Multiple (coordination-sensitive region), appears at a small number of cells near regime boundaries where no strict majority exists across seeds.

Four consistent patterns emerge across all phase diagrams. First, structural parameters dominate: retaliation capacity ( $\rho_{\text{max}}$ ), tariff capacity ( $\tau_{\text{max}}$ ), and export vulnerability ( $\psi_E$ ) are the primary determinants of regime, while follower depreciation capacity ( $d_{\text{max}}$ ) plays a secondary role. Second, perturbations can destabilise structure: introducing diplomatic cost ( $\xi > 0$ ) rapidly contracts the Deterrence region. Third, leader depreciation is a bounded non-monotone deterrence instrument. Fourth, coordination sensitivity is localised: Multiple cells appear primarily near zero-payoff contours. Results are

consistent across multiple seeds, with variation concentrated near regime boundaries.

In all phase diagrams, each cell shows the majority regime across five seeds. The black contour marks zero leader payoff, and star markers denote the LF, CENTER, and FF calibrations. Gray cells indicate Multiple outcomes, where no regime obtains a strict majority across seeds. Numeric annotations in these cells report the cross-seed mean leader payoff,  $\bar{L} = \frac{1}{S} \sum_{s=1}^S L^{(s)}$ , shown with sign and one decimal place.

### B. Core Structural Boundary

Fig. 1 maps follower retaliation ceiling ( $\rho_{\max}$ ) against leader tariff ceiling ( $\tau_{\max}$ ), revealing a clean monotone boundary separating Deterrence from Escalation. The boundary runs diagonally from ( $\rho_{\max} \approx 0.15$ ,  $\tau_{\max} \approx 0.10$ ) to ( $\rho_{\max} \approx 0.50$ ,  $\tau_{\max} \approx 0.42$ ). The regime is determined primarily by the ratio  $\rho_{\max}/\tau_{\max}$ : sustaining Deterrence against  $\rho_{\max} = 0.30$  requires  $\tau_{\max} \gtrsim 0.20$ , while at  $\rho_{\max} = 0.40$  the threshold rises to  $\tau_{\max} \gtrsim 0.30$ .

The Transition band spans roughly two grid cells throughout, forming a stable buffer between regimes. No Multiple cells appear at this resolution, indicating that the boundary is structurally decisive across seeds. Baseline stars fall as expected: LF lies in Deterrence, CENTER lies on the Transition boundary, and FF lies near Escalation. This ratio captures the relative strength of strategic instruments, determining whether retaliation can offset tariff-induced gains.

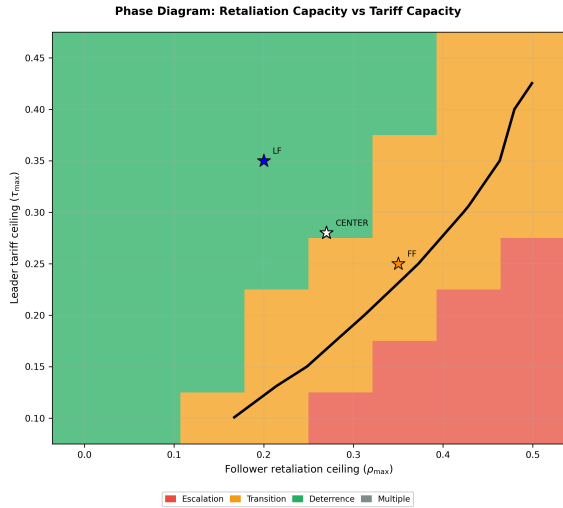


Fig. 1. Regime outcomes over follower retaliation and leader tariff capacity. The diagonal boundary shows that Deterrence and Escalation are separated primarily by the ratio  $\rho_{\max}/\tau_{\max}$ .

### C. Export Vulnerability Threshold

Fig. 2 isolates the effect of export sensitivity. A near-horizontal boundary separates Deterrence from Escalation at  $\psi_E \approx 1.75$ – $1.80$ , remaining flat across the full range of  $d_{\max}$ . Below this threshold, the leader sustains Deterrence regardless of follower depreciation capacity. Above it, Escalation dominates across all  $d_{\max}$ . Two Multiple cells appear at high

$d_{\max}$  ( $\geq 0.38$ ) and  $\psi_E \approx 2.0$ – $2.3$ , with mean payoffs of  $+0.9$  and  $+14.5$ , confirming localised ambiguity near the structural threshold. Export vulnerability is the primary determinant of regime, while follower depreciation capacity has limited independent effect.

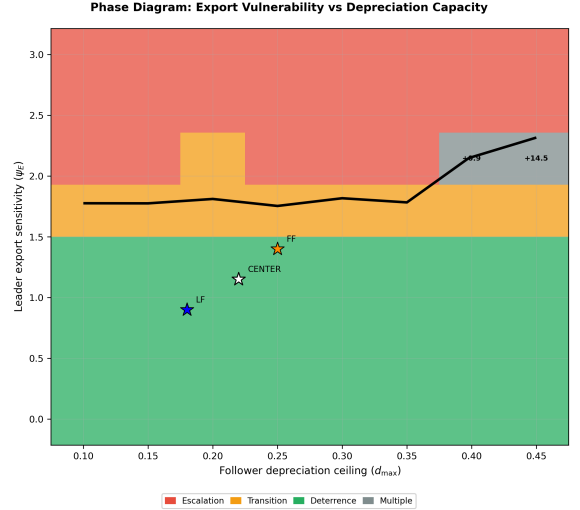


Fig. 2. Regime outcomes over follower depreciation capacity and leader export sensitivity. The boundary is nearly horizontal, with the escalation threshold at  $\psi_E \approx 1.75$ – $1.80$  remaining flat across the full range of  $d_{\max}$ , indicating that export vulnerability dominates depreciation capacity in determining regime outcomes.

### D. Follower Instrument Interaction

Fig. 3 shows that retaliation capacity dominates depreciation capacity. The zero-payoff contour runs nearly horizontally at  $\rho_{\max} \approx 0.23$ – $0.29$ , shifting by less than  $0.04$  as  $d_{\max}$  doubles. The phase diagram separates into three layers: Deterrence for

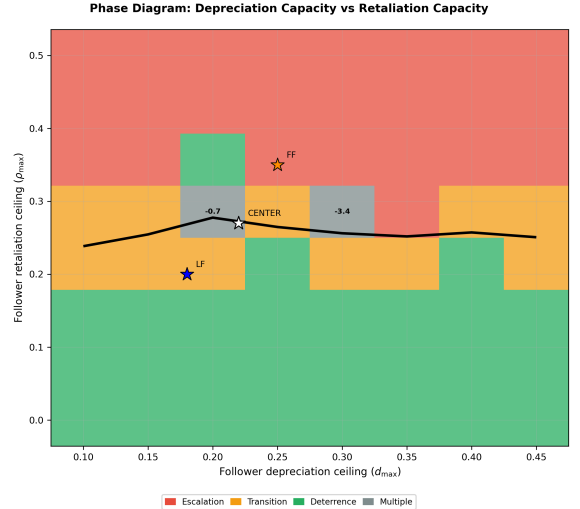


Fig. 3. Regime outcomes over follower depreciation capacity and retaliation capacity under the CENTER calibration. Deterrence occurs for  $\rho_{\max} < 0.18$ , Transition for  $0.18$ – $0.31$ , and Escalation for  $\rho_{\max} > 0.31$ , with only weak dependence on  $d_{\max}$ , indicating that retaliation capacity dominates depreciation capacity.

$\rho_{\max} < 0.18$ , Transition for  $\rho_{\max} \approx 0.18\text{--}0.31$ , and Escalation for  $\rho_{\max} > 0.31$ . Doubling  $d_{\max}$  shifts the escalation threshold by less than 0.02, indicating minimal substitutability between instruments.

Two Multiple cells appear near the Transition band at  $\rho_{\max} \approx 0.25\text{--}0.32$ , with mean leader payoffs of  $-0.7$  and  $-3.4$ , consistent with localised coordination sensitivity near the regime boundary.

#### E. Perturbation Analysis

All baseline results assume  $\xi = 0$ . Figs. 4 and 5 introduce diplomatic cost as a counterfactual perturbation.

In Fig. 4, the Deterrence boundary collapses sharply, decreasing from  $\rho_{\max} \approx 0.44$  at  $\xi = 0$  to  $\rho_{\max} \approx 0.07$  at  $\xi \approx 1.5$ . Even modest diplomatic cost eliminates Deterrence against moderate retaliation levels.

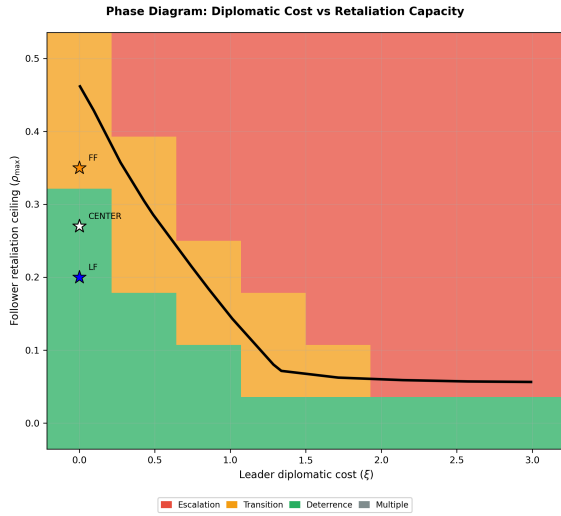


Fig. 4. Perturbation analysis over diplomatic cost and retaliation capacity. As  $\xi$  increases, the Deterrence boundary contracts sharply, with the critical threshold decreasing from  $\rho_{\max} \approx 0.44$  at  $\xi = 0$  to  $\rho_{\max} \approx 0.07$  by  $\xi \approx 1.5$ , indicating strong sensitivity of deterrence to diplomatic costs.

In Fig. 5, the joint effect of  $\xi$  and  $\psi_E$  is strongly amplifying: combinations such as  $\xi \approx 0.5$  and  $\psi_E \approx 1.0$  produce Escalation, even though each parameter alone would yield Transition or Deterrence. A leader that is both export-sensitive and diplomatically constrained faces the most contracted Deterrence region of any parameter combination examined.

#### F. Leader Depreciation Instrument

Fig. 6 introduces leader depreciation as a strategic instrument. The Deterrence boundary shifts upward from  $\rho_{\max} \approx 0.25\text{--}0.31$  at  $d_{L,\max} = 0$  to a peak of  $\rho_{\max} \approx 0.43$  at  $d_{L,\max} \approx 0.04$ , before dipping to  $\rho_{\max} \approx 0.31\text{--}0.33$  at intermediate levels and partially recovering thereafter. The resulting inverted-W contour reflects a non-monotone tradeoff between export gains and inflation costs. Five Multiple cells appear along the contour, with mean payoffs ranging from  $-6.3$  to  $+3.3$ . No Multiple cells appear deep inside the Deterrence region ( $\rho_{\max} < 0.20$ ). Baseline stars fall as expected: LF ( $d_{L,\max} = 0.15$ ,  $\rho_{\max} = 0.20$ ) lies in Deterrence;

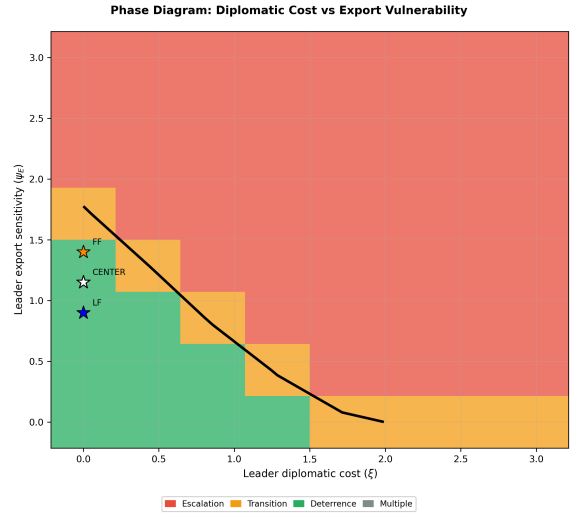


Fig. 5. Perturbation analysis over diplomatic cost and export sensitivity. Increasing  $\xi$  shifts the escalation boundary downward sharply, with the critical threshold falling from  $\psi_E \approx 1.8$  at  $\xi = 0$  to  $\psi_E \approx 0$  by  $\xi \approx 2$ , indicating that diplomatic costs and export vulnerability jointly amplify escalation risk.

CENTER ( $d_{L,\max} = 0.12$ ,  $\rho_{\max} = 0.27$ ) lies in Transition; FF ( $d_{L,\max} = 0.10$ ,  $\rho_{\max} = 0.35$ ) lies in Transition near the trough. Leader depreciation is a bounded deterrence instrument with an interior optimum.

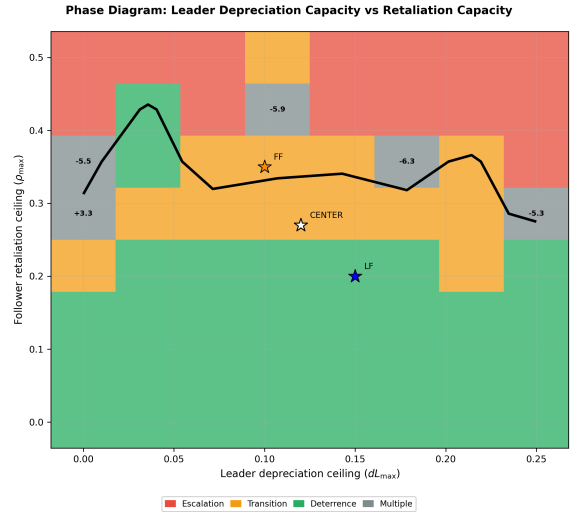


Fig. 6. Regime outcomes over leader depreciation capacity and retaliation capacity under the CENTER calibration. The zero-payoff boundary is non-monotonic in  $d_{L,\max}$ , forming an inverted-W shape: deterrence expands at low levels ( $d_{L,\max} \approx 0.03\text{--}0.05$ ) but contracts at higher levels ( $d_{L,\max} \gtrsim 0.10$ ), indicating that depreciation is a bounded instrument with diminishing and eventually adverse effects.

#### G. Convergence Dynamics

To connect phase structure with learning behaviour, we analyse policy trajectories using rolling-window averages of realised actions (Fig. 7). Under the LF calibration, the trajectory  $(\rho(t), \tau(t))$  exhibits rapid early movement followed by damped oscillations and eventual stabilisation, converging to

a stable point at  $(\rho \approx 0.125, \tau \approx 0.345)$ . Leader actions converge to a near-pure strategy, while follower retaliation remains weakly mixed near equilibrium, reflecting multiple near-optimal responses. This is consistent with the payoff structure in Section II: since  $\rho$  provides no direct follower export benefit, the follower is structurally near-indifferent across retaliation levels at equilibrium. Trajectories remain confined within a single regime region throughout training.

Convergence is further examined using a best-response (BR) probe. After training, the leader policy is frozen and a fresh follower agent is run to identify an approximate myopic best response. The resulting payoff changes remain modest relative to equilibrium payoffs and do not qualitatively alter the learned policy regime. These results are consistent with approximate best-response behaviour, indicating that the learned policies represent locally stable co-adaptive strategies rather than transient artifacts.

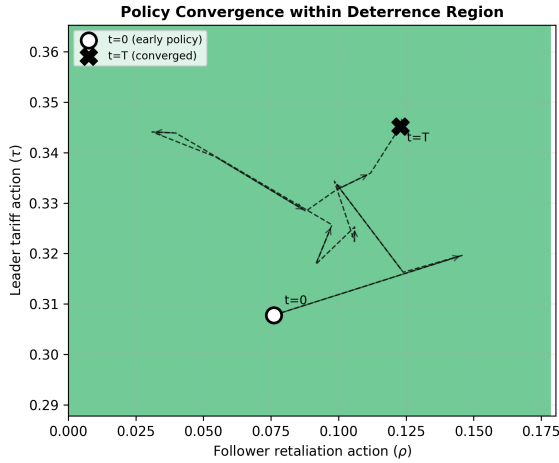


Fig. 7. Rolling-window policy trajectory under the LF calibration. The trajectory converges to a stable point within the Deterrence region, supporting a basin-selection interpretation of learning dynamics.

#### H. Coordination Sensitivity and Initialization Effects

Multiple cells appear primarily near zero-payoff contours across all phase diagrams, with mean payoffs typically within the range  $[-6.3, +14.5]$ . These regions reflect coordination sensitivity in the learning dynamics, where identical structural parameters produce different stable policy outcomes depending on initial conditions and exploration paths. Within each seed, agents converge to a stable policy. Disagreement across seeds arises because the system admits multiple stable policy attractors near the regime boundary, and the outcome depends on which attractor is reached during early exploration, not on noise in the learning process.

Lower Q-initialisation expands these regions, indicating that initialisation acts as an equilibrium selection mechanism. The overall phase boundary remains stable across initialisations, confirming that regime separation is structurally determined and that coordination sensitivity is a boundary phenomenon rather than a property of regime interiors. This reflects sensi-

tivity in attractor selection rather than instability in the learning process.

### V. DISCUSSION

#### A. Economic Interpretation

The phase diagram results offer a structural perspective on trade conflict that complements both static game-theoretic analysis and empirical studies of specific episodes. The central finding, that the Deterrence–Escalation boundary closely tracks the ratio of follower retaliation capacity to leader tariff space, has a clear economic interpretation. A leader whose tariff instrument is constrained, whether by WTO commitments, supply chain exposure, or domestic political costs, faces a structural disadvantage when the follower retains credible retaliation capacity. The conflict outcome is determined not by any single policy choice but by the underlying asymmetry in instrument ranges. While stylized, the model isolates strategic interactions between policy instruments, allowing structural mechanisms to be examined independently of institutional complexity.

The depreciation finding adds nuance to this picture. Currency depreciation can extend the Deterrence region, but only at moderate levels. Beyond an interior optimum, the inflationary costs of aggressive depreciation erode the leader’s payoff faster than the export competitiveness gains accumulate. This non-monotone relationship suggests that currency intervention is most effective as a complement to tariff policy rather than a substitute for it. The result connects to the broader currency wars literature [6], providing a phase-diagram account of when competitive depreciation stabilises and when it destabilises.

The perturbation analysis reveals a compounding vulnerability. Diplomatic friction and export sensitivity are individually destabilising, but their combined effect is disproportionately destabilising, with the combined impact larger than would be expected from either factor alone. A leader facing both constraints simultaneously, high export exposure and non-trivial diplomatic costs, operates in the most contracted Deterrence region of any configuration examined.

#### B. Methodological Implications

The phase diagram methodology offers a reusable tool for analysing parameter-dependent regime structure in strategic settings. Unlike static equilibrium analysis, the phase diagram reveals regions where multiple stable outcomes coexist, and outcome selection depends on the learning dynamics. Rather than characterising a single equilibrium, it maps the full parameter space to outcome regions, revealing not just where stable outcomes arise but how robust they are to structural variation. This approach is applicable beyond trade conflict to any repeated game where the outcome type depends sensitively on structural parameters.

The coordination sensitivity finding illustrates a distinct contribution of MARL to economic analysis. Standard equilibrium analysis identifies Nash equilibria but does not predict which outcome is selected when multiple stable outcomes exist. The Q-learning framework surfaces this selection problem



empirically: at boundary cells, the exploration path during early learning determines which attractor the system converges to. This reflects a substantive coordination problem rather than a limitation of the method.

## VI. LIMITATIONS

**Learning protocol.** The model implements simultaneous Q-learning rather than true sequential Stackelberg play. In each period, both agents update their policies concurrently, so the leader does not have a formal first-mover commitment advantage beyond what emerges from the asymmetric payoff structure and state representation. The resulting equilibria are more appropriately interpreted as co-adaptive stable policies rather than as Stackelberg equilibria in the classical sense.

**Discretised action and state spaces.** Both agents operate over binned action grids, which limits the precision of policies near equilibrium. Continuous-action methods such as deep RL or actor-critic architectures would allow finer resolution, but at the cost of the interpretability and stable convergence behaviour observed in this setting. The sensitivity of Multiple-cell boundaries to bin resolution is an avenue for future work. Increasing the number of seeds or refining the discretization may sharpen boundary estimates, but does not alter the qualitative regime structure observed across all parameter configurations.

**Scalability.** Tabular Q-learning scales poorly to large state-action spaces. The current architecture is tractable because the economic model is deliberately parsimonious, but extending to multi-sector trade, more than two countries, or richer instrument sets would likely require function approximation methods. Tabular methods are sufficient given the low-dimensional state space and enable direct interpretation of learned value functions.

**Trade model simplification.** The model assumes a single composite good, a two-country setting with asymmetric roles, and zeroed cross-elasticities to isolate primary strategic channels. Real trade conflicts involve heterogeneous sectors, asymmetric country sizes, and complex supply chain interdependencies that the current framework does not capture. Calibration to empirical elasticity ranges helps ground the model in observed behaviour, but the framework is best understood as a stylised representation of strategic structure rather than a quantitative forecasting tool.

**Behavioural parameters.** Inequity aversion, spite, and diplomatic cost parameters are set to zero in the structural baseline. Perturbation analysis in Section IV-E provides a partial sensitivity check, but a fuller exploration of behavioural effects is left for future work.

Quantifying uncertainty in boundary location through bootstrap resampling or increased seed counts is a natural extension for future work.

## VII. CONCLUSION

We model a two-country tariff conflict as an adaptive strategic interaction and simulate it using tabular Q-learning across a structured parameter grid. Three findings emerge.

First, the Deterrence–Escalation boundary closely tracks the ratio of follower retaliation capacity to leader tariff space, providing a compact structural characterization of conflict outcomes. Second, moderate leader currency depreciation shifts the Deterrence boundary, but the effect is non-monotonic, with an interior optimum beyond which inflation costs outweigh the associated export gains. Third, near structural boundaries, identical parameters produce divergent outcomes across learning runs, revealing coordination-sensitive regions where early exploration influences which stable policy regime is realised.

Together, these results show that phase diagram methods combined with multi-agent reinforcement learning provide a tractable and interpretable framework for characterising regime structure in strategic economic settings. The approach extends naturally to other repeated games where outcome type depends sensitively on structural parameters.

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